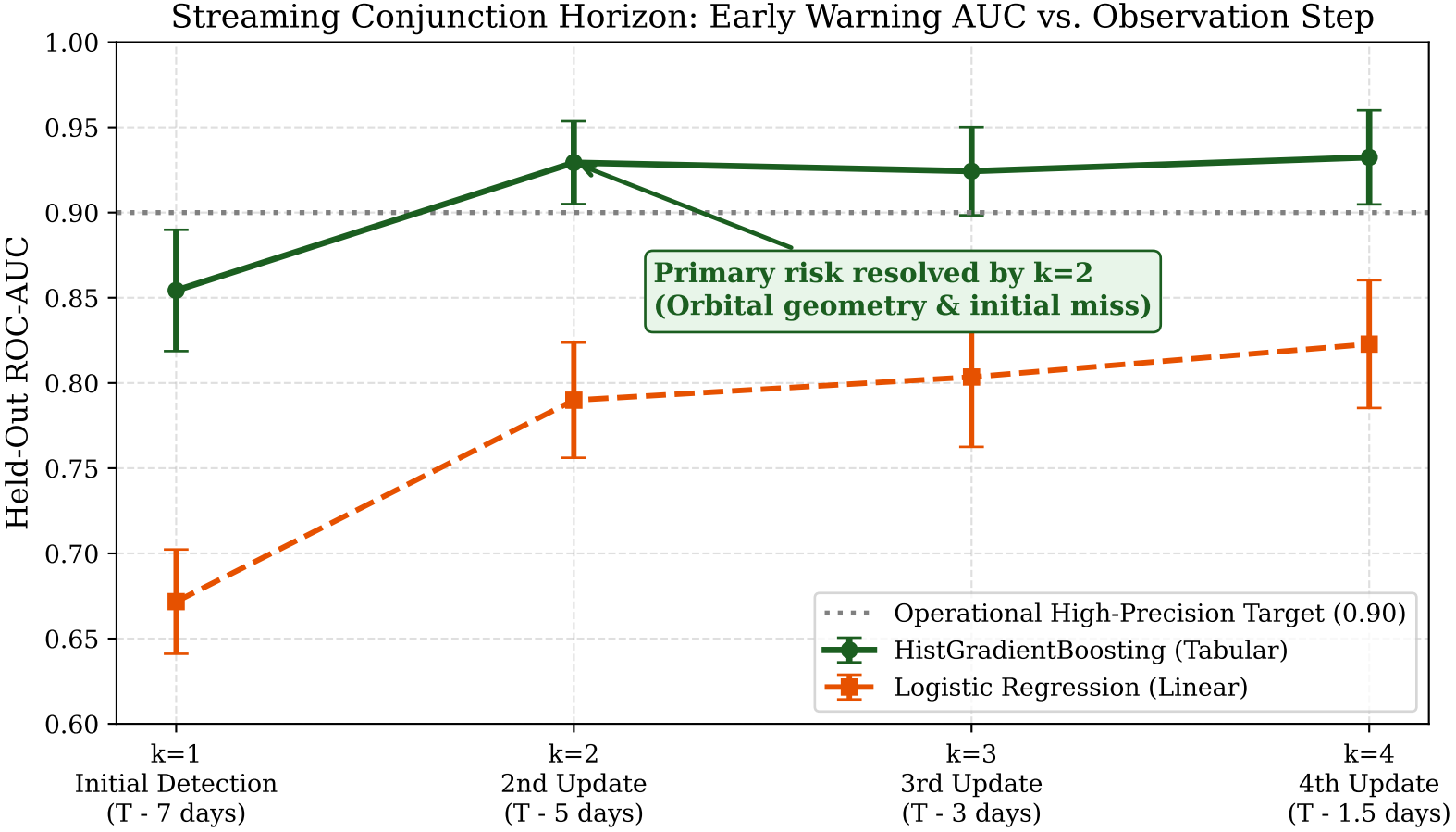
Streaming Conjunction Horizon: Early Warning AUC vs. Observation Step
Held-Out ROC-AUC
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
Primary risk resolved by k=2
(Orbital geometry & initial miss)
Operational High-Precision Target (0.90)
HistGradientBoosting (Tabular)
Logistic Regression (Linear)
k=1
Initial Detection
(T - 7 days)
k=2
2nd Update
(T - 5 days)
k=3
3rd Update
(T - 3 days)
k=4
4th Update
(T - 1.5 days)